

Additional Help and Resources

R and RStudio Resources

R

- [Download: R from CRAN](#)
 - This is where you can download the R language software for FREE for your own computer.
 - Choose your operating system (Mac OS or Windows or Linux/Unix)
 - NOTE: For Windows, you should also download and install [Rtools](#) - this is technically optional, but is useful to have. Make sure to download the one for your R version.
- [R Cookbook](#)

RStudio

- [Download: RStudio IDE Desktop](#)
 - Note: Windows is listed at the top - just scroll down to see the installer for the Mac OS as well. There are also installers for the versions of Linux/Unix.
- [RStudio Education](#)
- [RStudio Cloud Tutorials](#)
- **** Quick-R ****

Crosswalking from SAS to R

- [The sassy package](#) - “making R easy for SAS programmers”
- [SAS-to-R Cheatsheet](#)
- [SAS-vs-R in Pharma Cheatsheet](#)
- [R for SAS Users Tutorial](#)
- [PROC R to run R code in SAS](#)

- [Posit Blog 2013 - Learn R as a SAS Users](#)
- [Writing SAS MACROs in R - PharmaSUG 2025 paper](#)

R Graphics

- [Book: ggplot2](#)
- [R Graphics Cookbook - online book](#)
- [Cookbook for R \(graphics\) - earlier version](#)
- [R Graph Gallery](#)
- [Book: R Graphics, 3rd edition by Paul Murrell and accompanying R Graphics - book website](#)
- [R Charts Website](#)
- [Book: Interactive web-based data visualization with R, plotly, and shiny](#)
- [Book: R For Data Science - Layers](#)
- [Book: The Epidemiologist R Handbook - Data Visualization Section](#)

R Packages for Tables

Making Tables (*without Rmarkdown*)

- The `table()` function, run `help(table, package = "base")`
- [gtsummary](#)
- [gmodels](#), see `CrossTable()` function

Rmarkdown Tables

- [gt](#)
- [gmodels](#)
- [gtsummary](#)
- [arsenal](#)
- [huxtable](#)
- [flextable](#)

Tables with Graphics

- [R Gallery Tables Summary](#)
- [gtExtras](#)
- [skimr](#)

RMarkdown Resources

- [Rmarkdown Tutorial](#)
- [Book: R Markdown: The Definitive Guide](#)
- [Book: R Markdown Cookbook](#)

Quarto Resources

- [Quarto](#)
- [Quarto User Guide](#)
- [Chapter 28 Quarto in “R for Data Science \(2e\)”](#)

Online Training and Courses

- [Code Academy](#)
- [Software Carpentry](#)
- [swiRl - Learn R in R](#)
- [Datacamp](#)
 - [R for SAS Users - My Datacamp Course](#)
- [Coursera](#)
 - [Reproducible Templates for Analysis and Dissemination - My Coursera Course](#)
- [My Courses at Emory:](#)
 - [Emory N741](#)
 - [Emory N736](#)

More Helpful Online Books on R and Statistics with R

- [Book: SAS and R: Data Management, Statistical Analysis, and Graphics \(second edition\)](#)
- [Book: R for SAS and SPSS Users \(Statistics and Computing\) - on Amazon and FREE Online Version](#)
- [Book: Statistical Inference via Data Science](#)
- [Book: The Epidemiologist R Handbook](#)
- [Book/Course: Stat 545](#)
- [Book: Statistical Inference via Data Science: A ModernDive into R and the Tidyverse](#)
- [Book: R in Action](#)
- [OpenIntro Statistics](#)
- [Mastering Software Development in R](#)
- [UCLA's Office of Advanced Research Computing \(OARC\) - R Software Resources](#)
- [PharmaSUG 2022 - Paper AP-193: “R syntax for SAS programmers”](#)

Other places to get HELP

- [RStudio/Posit and Contributed Cheatsheets](#)
- [StackOverflow](#)
 - I encourage you to create an account so you can post questions. But even without an account you can search for and find answers to your questions and error messages.
- [Google](#)
 - You can often cut and paste error messages in Google to find answers - most likely will redirect you to Stack Overflow.
- Package vignettes for packages on CRAN
 - Here is [one vignette for dplyr](#)
 - These will often help you get started.
- Github package issues
 - Many packages will host their code on Github which includes an “issues” tab. This can be a good place to see what other problems people may be having with a given package.
 - [dplyr issues on Github](#)
- CRAN package site
 - [dplyr on CRAN](#) - spend time looking at:
 - * the [README](#) for the package or
 - * [bug reports](#) or
 - * [NEWS](#) which will detail the changes for each version updates
- [R Bloggers](#)
 - This is a really good website which curates thousands of people who are R developers, users and programmers who post articles about R.
- [STHDA Website for “Statistical tools for high-throughput data analysis”](#)
 - This website will often come up when “Googling” for answers. It has a lot of ads but often has very helpful examples.
- [Quick-R](#)
 - This website was originally created by Robert I. Kabacoff, who wrote the [Book: R in Action](#).
 - However, the website has now been taken over by [Datacamp](#).